Course overview

Modeling the Distribution of Biodiversity under Global Change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Scope

Provide students **training in Species Distribution Modeling** to infer and discuss the **interactions** and potential **impacts of global climate changes on biodiversity**.

It is mostly **hands-on oriented**, with a strong component on:

1) biodiversity and climatic **data acquisition**, **management** and **visualization** (e.g., species occurrence data and the new Shared Socioeconomic Pathway scenarios of climate change);

2) **distribution modeling** frameworks using state of the art algorithms (e.g., machine learning extreme gradient boosting).

Structure

Part 1 : Lectures and hands-on oriented coding in R computing language.

Part 2 : Written assignment, which is an individual research study addressing:

(1) the interactions between, or

(2) the impacts of, global climate changes on biodiversity.

Part 3 : The SDM summit: oral presentations based on the individual assignments + Q/A.

Sessions (theory)

Session 1: Foundations of Marine Macroecology and Niche Theory

Session 2: Spatial foundations, data acquisition and quality

Session 3: The Architecture of Species Distribution Models

Session 4: Evaluating, Transferring, and Ensembling Species Distribution Models

Session 5: Real-World Applications & Global Change Projections

Sessions (practice)

P1 Introduction to R and RStudio

P2 R as a Geographic Information System

P3 Downloading and managing biodiversity data in R

P4 Downloading and managing environmental data in R

P5 Developing Species Distribution Models in R

P6 Performance of Species Distribution Models

P7 Data-driven hyperparameter and monotonic constrains

P8 The diversity of algorithms and ensembles

P9: Niche overlap estimates.

Assessment methods and criteria

Written assignment

(60% of the grade A-F; to be submitted on **November 16th 2pm**).

Oral presentation

(40% of the grade A-F; to be held on **November 16th 9am**).

Resources

https://github.com/jorgeassis/courseModellingDistributionBiodiversity

Sessions (PDFs), Data, Code, Challenges, Solutions and Recipes.

Relevant questions

[book a meeting] jorgemfa@gmail.com